

Report

**Project Seminar Biomedical Image Analysis
Advanced Practical Deep Learning for Biomedical Image Analysis**

Heidelberg University

Polyps Segmentation in Colonoscopy

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Project Seminar Biomedical Image Analysis
Advanced Practical Deep Learning for Biomedical Image Analysis

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Abstract

Colorectal Cancer (CRC) is among the most common cancers globally. They develop from adenomatous polyps that are easy to detect and remove during colonoscopy. Early and accurate detection of polyps can improve the patient's outcome. However, polyps can be missed during colonoscopy because of smaller size and low-texture contrast. Therefore, minimizing the rate of missed detections is a critical goal.

This paper investigates the implementation of the SwinE-Net, a hybrid transformer-based architecture that uses the Swin Transformer to detect local contextual features while Convolutional Neural Network (CNN) based Efficient-Net detects the global features of the polyps from the colonoscopy images.

Experimental results demonstrate that SwinE-Net achieves up to 0.938 mean Dice coefficient (mDice) and 0.892 mean Intersection over Union (mIoU) on seen datasets such as Kvasir-SEG and CVC-ClinicDB. Furthermore, the model generalizes well, achieving up to 0.906 mDice and 0.842 mIoU on unseen datasets including ColonDB, ETIS-Larib, and EndoScene.

Keywords: Colonoscopy, Polyps Segmentation, Swin Transformer, Efficient Network, Convolutional Neural Network.

Chapter 1

Introduction

In this chapter, we give an overview of this report. Section 1.1 explains the motivation for automated polyp segmentation in colonoscopy. Section 1.2 states the goals of this work. Section 1.3 outlines the structure of the report.

1.1 Motivation

Colorectal cancer (CRC) is one of the leading causes of cancer-related death worldwide; early detection and removal of polyps during colonoscopy can reduce the incidence of CRC by up to 90 %. However, manual inspection has a missed rate of 20–30 % in routine practice, due to factors such as operator fatigue, poor visibility, and subtle appearance of the lesion. Automating polyps segmentation, that is, drawing precise masks around suspected lesions, promises to (1) alert the endoscopist in real time, (2) reduce false negatives, and (3) standardize reporting between practitioners. Despite advances in convolutional neural networks (CNNs) and vision transformers, most existing methods either lack global context or fail to preserve fine details, motivating a hybrid approach.

1.2 Goals

This report describes our work to reproduce and analyze the recent paper: “SwinE-Net: Hybrid deep-learning approach to novel polyp segmentation using convolutional neural network and Swin Transformer.” by Park et.al [1]. In the paper, the authors propose a new segmentation model, SwinE-Net, that combines an EfficientNet backbone (for fine details) with a Swin Transformer (for global context). They add multi-dilation convolutions, feature-fusion blocks, and attention-based decoding to produce highly accurate polyp masks. Our objective is to:

- **Implementation:** Build from scratch the exact SwinE-Net architecture in PyTorch, following their descriptions of each module (dual encoders, MDC, MFA, CBAM, Inception-Res decoder, loss functions).
- **Evaluation:** Train in the public Kvasir-SEG and ClinicDB datasets, then test on unseen benchmarks (if time permits) to verify their reported improvements in Dice and IoU. We will also conduct ablation studies to confirm which components drive performance.

1.3 Overview

Chapter 2 surveys related work from U-Net variants and attention-augmented models to pure transformer approaches in polyp detection and other medical domains.

Chapter 3 describes the design and implementation of our method, including the handling of the data set, the model architecture (MDC, MFA, CBAM, decoder) and the loss formulation.

Chapter 4 details the code structure, the training pipeline, the hyperparameter choices, and the engineering optimizations.

Chapter 5 presents experimental results: quantitative comparisons (Dice and IoU tables), qualitative examples, and ablation studies evaluating each architectural component.

Chapter 6 discusses the findings: generalization performance, limitations (data size, boundary precision), and lessons learned.

Chapter 2

Related Work

Chapter 3

Fundamentals

3.1 Overview and General Concept

3.2 Software System

Chapter 4

Methodology

Here we define methodology

Chapter 5

Experimental Results

Chapter 6

Discussion and Future Work

Appendix A

Project Timetable Comparision

Table A.1: Projected Timeline and Task Allocation

Week	Pranjal	Hanisha
Week 0	PPT-1	PPT-1
Week 1 – Kickoff	Literature Review	Understand SwinE-Net architecture
Week 2 – Dataset Collection and Preprocessing	Gather Dataset	Dataset Cleaning
Week 3 – CNN and Transformer Model Setup	Implement E-Net based CNN feature extractor	Implement Swin Transformer
Week 4 – Architecture Development	Integrate E-Net and Swin Transformer	Implement multidilation convolution block and multifeatured aggregation block
Week 5 – Decoder and Attention Mechanisms	Implement attentive deconvolution network	Integrate and test CBAM (Convolutional Block Attention Module)
Week 6 – Loss Function and Training Setup	Implement the loss function, Prepare PPT-2	Set up the training pipeline, Prepare PPT-2
Week 7 – Training the Model	PPT-2, Train and adjust the model	Conduct initial testing and evaluate early performance
Week 8 – Evaluation and Fine Tuning	Evaluate model performance on unseen datasets (Maybe not)	Fine Tuning
Week 9 – Verification and Report Creation	Verify results, clean git, verify documentation	Finalize report
Week 10	Project Presentation-3 and Report Submission	

This is not complete

Table A.2: Actual Timeline and Task Allocation

Week	Pranjal	Hanisha
Week 0	PPT-1	PPT-1
Week 1 – Kickoff	Literature Review	Understand SwinE-Net architecture
Week 2 – Dataset Collection and Preprocessing	Gather Dataset	Dataset Cleaning
Week 3 – CNN and Transformer Model Setup	Implement E-Net based CNN feature extractor	Implement Swin Transformer
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Week 10	Project Presentation-3 and Report Submission	

Bibliography

- [1] K.-B. Park and J. Y. Lee, “Swine-net: Hybrid deep learning approach to novel polyp segmentation using convolutional neural network and swin transformer,” *Journal of Computational Design and Engineering*, vol. 9, pp. 616–632, 04 2022.